

Activity: logistic regression

Group members:

Medical school admissions

The estimated logistic regression model for the relationship between GPA and acceptance to medical school is

$$\log\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA},$$

where $\hat{\pi}$ is the estimated probability of acceptance.

1. What is the estimated odds of acceptance for a student with a GPA of 4.0?
2. What is the estimated *probability* of acceptance for a student with a GPA of 4.0?
3. Calculate the odds ratio comparing odds of acceptance for a student with a GPA of 4.0 to a student with a GPA of 3.0.
4. Calculate the odds ratio comparing odds of acceptance for a student with a GPA of 3.8 to a student with a GPA of 2.8.